

A decorative graphic on the right side of the slide features several circular and semi-circular frames of varying sizes. These frames contain images of modern skyscrapers with glass facades, viewed from low angles looking up. The frames are arranged in a vertical column, with some overlapping. The background of the entire slide is a solid dark blue.

Linux Primer: Basics



Contents

1. Get around Linux
2. Use basic file commands



Get around Linux



Logging on & Who are you

Using an SSH Client we will log on to Linux. Who are you logged in as?

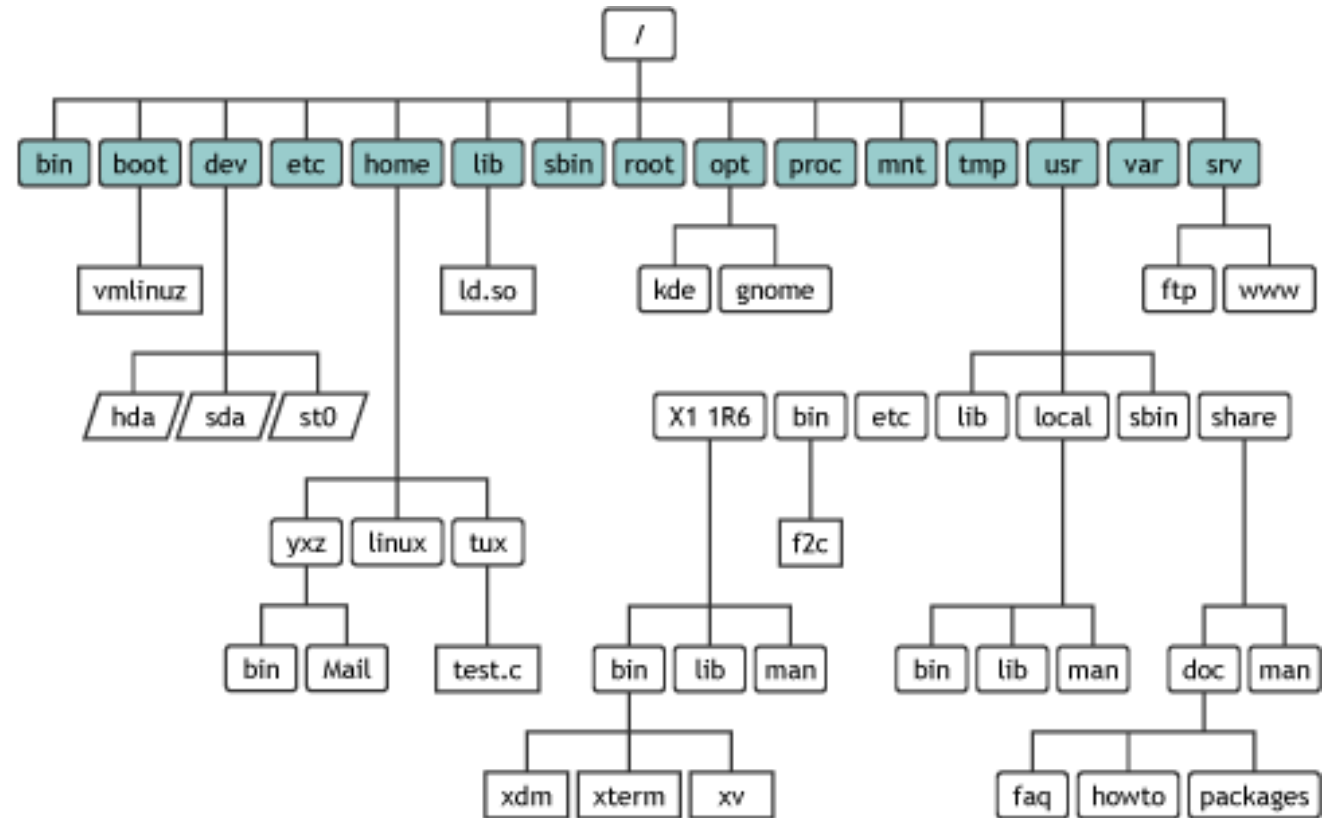
```
ssh me@somelinuxsystem.com
```



Logging on & Who are you - Overview

The Linux file system in short; a quick tour

- /tmp, /home, /usr, /var what are they?
- Names for directories like /home, /home/steve, / and /root



See: https://en.wikipedia.org/wiki/Filesystem_Hierarchy_Standard



Getting Help

There are a few places you can go to for help.

Man pages are built into Linux so you can use the man command:

- `man ls` - **man** `ls`
- `man cp` - **man** `cp`

Stackoverflow.com:

- Some useful answers, but not always the simplest, or correct so you should try them out
- When trying out destructive commands make sure you create a directory with some test files in

When searching the web, also include the operating system type of RHEL or Debian.



Use basic file commands



Changing Directory

Using CD to get about and some useful short cuts.

- Absolute and relative paths
- tilde (~) and minus (-)

```
# All go home!  
cd ~  
cd  
cd $HOME  
  
# change into specific folders  
cd /home/steve  
cd /etc/bin
```



Creating directories

- Creating directories with `mkdir`
- Creating child directories when the parent doesn't exist

```
mkdir newfolder  
cd newfolder
```



TASK 1

1. Search the web to find out how to create child directories using mkdir if the parent does not exist
2. Once you have found the command create the directory structure
 - `steve/nick/frank`



Viewing File Attributes

Viewing the attributes of files and directories with ls:

- Core options: -l, -a, -ltr, -S, -R.

What do all those columns mean?

```
ls somefolder
ls -a somefolder
ls -ltr somefolder
ls -S somefolder
ls -R somefolder
```



TASK 2

1. List attributes of all files and directories in `/etc/cron.daily`
2. List the file/directories in `/etc/cron.daily` in ascending order of size
3. Show all hidden files in your home directory
4. How do I change the output of the `-l` option to show a more readable file size? e.g. MB, GB, etc. List the `/etc/cron.daily` directory using the option you find

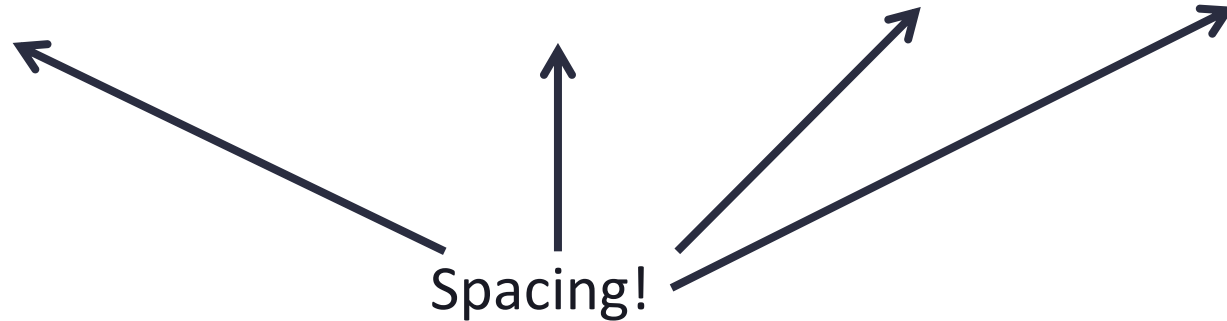


Command Syntax

All commands are lowercase. Spacing is important:

- Between command and options
- Between options and arguments
- Between all arguments

command [-options] [arg1 arg2 argn]



References

- <https://www.tecmint.com/cd-command-in-linux/>
- <https://www.tecmint.com/15-basic-ls-command-examples-in-linux/>
- <https://shapeshed.com/unix-ls/>





Basics Exercise (10 minutes)

1. List the files and directories in /etc
2. List the files and directories in your home directory
3. List all the files and directories in your home directory
4. Recursively list the /usr/include directory
5. Change to the following locations using absolute path
 - /usr/bin
 - /usr/local/bin
 - /var/log
6. Change to the following locations using relative paths, starting from /var/log and then relative from the next, and so on
 - /etc/rc.d
 - /usr/share
 - /usr/lib





Solutions

On the next slides you will find solutions to:

- Task 1
- Task 2
- The exercise



Task 1 solution

Question: How to create child directories using mkdir if the parent does not exist?

Answer: You can use mkdir with the -p flag to create parent directories as needed.

```
mkdir -p steve/nick/frank
```



Task 2 solution

Question: How to list attributes of all files and directories in /etc/cron.daily?

Answer: The `ls` command with the `-l` flag lists all files and directories with their attributes.

```
ls -l /etc/cron.daily
```

Question: How to list the files/directories in /etc/cron.daily in ascending order of size?

Answer: The `ls` command with the `-l` and `-S` flags sorts by file size. The `-r` flag reverses the order to ascending.

```
ls -lrS /etc/cron.daily
```



Task 2 solution (cont.)

Question: How to show all hidden files in your home directory?

Answer: The `ls` command with the `-a` flag shows all files, including hidden files.

```
ls -a ~
```

Question: How to change the output of the `-l` option to show a more readable file size? e.g. MB, GB, etc. List the `/etc/cron.daily` directory using the option you find.

Answer: The `ls` command with the `-lh` flag shows file size in human readable form.

```
ls -lh /etc/cron.daily
```



The Exercise 1 of 4

Question: How to list the files and directories in /etc?

Answer:

```
ls /etc
```

Question: How to list the files and directories in your home directory?

Answer:

```
ls ~
```



The Exercise 2 of 4

Question: How to list all the files and directories in your home directory?

Answer: The `ls` command with the `-a` flag shows all files, including hidden ones.

```
ls -a ~
```

Question: How to recursively list the `/usr/include` directory?

Answer: The `ls` command with the `-R` flag lists files and directories recursively.

```
ls -R /usr/include
```



The Exercise 3 of 4

Question: How to change to the following locations using absolute path?

Answer:

cd /usr/bin

cd /usr/local/bin

cd /var/log



The Exercise 4 of 4

Question: How to change to the following locations using relative paths, starting from /var/log and then relative from the next, and so on?

Answer: Using relative paths is dependent on the current directory. If starting from /var/log:

```
cd ..
```

```
cd etc/rc.d
```

```
cd ../..
```

```
cd usr/share
```

```
cd ../lib
```



What we covered

- Get around Linux
- Use basic file commands



Thank You

Do you have any questions?

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